Foundation for Virtual and Augmented Reality Systems

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Week – 05

Lecture – 25

Lec-25: Human Factors in XR

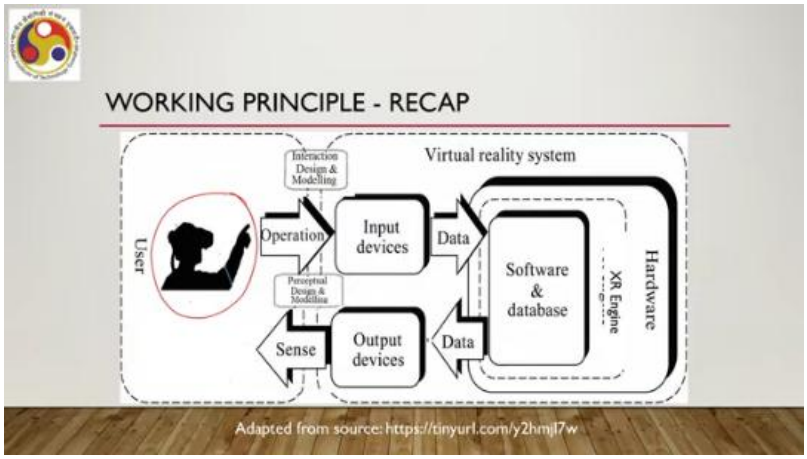
Hello and welcome to the NPTEL MOOC's course on Foundations for Virtual and Augmented Reality Systems. We are going to start our discussion on a new topic, topic number 13, which is on Human Factors in Extended Reality. Before we start, let us quickly recap our learning so far. So, we talked about the basic idea of XR, Extended Reality. issues and challenges.

Then we talked about the hardware and the software that are typically used to build, develop, and deploy XR systems. And in the previous lecture, we also started talking about the idea of interactive systems, human factors, and their relation to XR. So, we just gave some introductory ideas. The primary concern in the previous lecture was on the idea of interactive systems.

Since in extended reality, we put emphasis on interaction with the digital content. So, XR broadly belongs to the interactive systems domain. and the associated issues and challenges are to be addressed while we are trying to build and deploy an XR system. So, that was a crucial concept in understanding the foundation of extended reality systems, and in the previous lecture we started our discussion on that. We are going to continue our discussion on the topic.

of interactive systems and their relation to XR. In particular, in this lecture, we are going to talk about human factors. Earlier, we emphasized the role of human factors in interactive system development. Here, what we will do is we will try to understand in a little bit more detail the human factors that are relevant for XR or extended reality and how the knowledge helps a developer build better XR systems.

Now, before we start, let us quickly recap the working principle of the extended reality systems. On one side we have the user, and on the other side we have the system. Now, the system we can view as comprising the hardware the middleware, which we are terming as the XR engine, and the software or the application software. The user and the system interact through input devices.



So, the user provides input, and the system provides output to the user through output devices. The output that is generated is to be sensed by the user. Now, two things are important here. One is interaction design; the other one is perceptual design. So, interaction design is important to give the user the ease of interacting with the system.

Earlier we talked about such ease of use as usability. So, interaction design is very, very important to build a usable system which the user can interact with easily and the output is also generated for the user to be appropriately sensed, understood, or perceived. So, from the point of view of usability both interaction design and modeling as well as perceptual design and modeling. One related to the input, the other related to the output are very very important.

The interaction and the perceptual stimuli that is the output generated by the system are supposed to be designed to create the best possible illusion of reality. Remember that our goal is to make a system that enhances presence. When I say it enhances presence, what I mean is that it gives us a sense of reality, as if we are in the real world, so the interaction and perceptual stimuli are supposed to be designed to create that best illusion, best possible illusion of reality. It is not reality, but we are creating an illusion of reality.

Now this idea of course as we have mentioned earlier is very crucial for VR because here we are supposed to be totally immersed in the virtual environment to give an immersive

experience. Since in case of AR or MR systems immersion is partial we are still able to view or experience the real world so this illusion may not be very significant but still not totally insignificant also. So, we have to think about how to create the best possible illusion of reality to give a better XR experience. So, that is our objective to understand how to create that illusion.

In order to understand that, we need to understand how our senses, how our perception works and what is the physiology that drives our perception. So, let us try to delve a little deeper into our perceptual system. By now, it must have been clear to you the role of perception in our experience working with any interactive systems in general and extended reality systems in particular.

Since the illusion of reality is primarily applicable to VR systems, we will mostly refer to VR systems while continuing our discussion. Now, VR refers to an immersive experience that by now we have learned about. VR systems are supposed to be designed to have interactions that let us feel immersed. Now, this feeling is nothing but perceiving that we are immersed.

But one important thing here to note is that by doing so, we are trying to fool ourselves. It is a kind of fooling ourselves into believing that we are in a different world. So, essentially what it tells is that we are trying to perceive the system as if we are in our regular, familiar system. So, we are trying to fool our perception, perceptual process, perceptual system to make us believe that we are still in our familiar system.

Unless we are able to do that, complete immersion, complete presence will not occur. So, our objective is to have as much presence as possible. In order to do that, our ability to fool ourselves should be maximum. The fact is that our bodies are not really designed for virtual reality; they are designed to perceive an artificial world, a virtual world.

So, our bodies are designed, or our body organs are designed, to perceive the world around us, which we call the real world. So, they are not designed to perceive a virtual world. So, by applying artificial stimulation to our senses, what we are trying to do is affect the biological mechanisms that have evolved over a very, very long period of time in a natural environment. So, our body gets trained to perceive the real world based on not only sensory organs but also experience, as we shall soon see.

How the physiology and the experience stored in long-term memory combine together to give the idea of perception, but when we are applying artificial stimulation, we are affecting

that biological mechanism that has been learned over a long period of time, and how well we are able to fool that system decides how good the overall virtual reality experience will be. In fact, by trying to do so, we are also providing input to our brain that is not consistent with our real-life experiences. Most of the time, we will be doing things to experience the virtual world that are generally not consistent with the way we do things in the real world given the same situation.

So, generally we are sending signals that are not consistent with our real-life behavior. So, if I try to do that, if we try to do that, what can be the potential outcomes of trying to do something that our body is not designed to process, that our brain is not designed to process? The potential outcomes can be There can be many potential outcomes.

One is our bodies adapt to the new stimuli, making us unaware of flaws if any in the system. That means we are adapting to the system. We believe that the system is actually the real world rather than it being a virtual artificial system. Other possible outcomes can be that our bodies may not be able to adapt So, the body may react with increased fatigue or headaches.

So, some adverse effects can be seen on the body when we are trying to fool ourselves into believing that we are in the real world while actually we are in the virtual world. Now, this non-adaptation, as we have discussed earlier in one of the previous lectures, may lead to cyber sickness that is dizziness, nausea or a large set of different symptoms that we can experience while experiencing virtual reality systems. To design a good system or to understand flaws of a given system, we should have some understanding of how our perceptual system works.

That is the crux of the matter. So, we need to understand human perception for two purposes. One is to design a better system or to understand the flaws of a current system or a given system and improve or resolve those shortcomings. So let us try to understand the human perceptual system. Now there is a term called perceptual psychology.

Many of you may have heard of this term. It is the science of understanding how the brain converts sensory stimulation into perceived phenomena. So we have sensors; signals from the real world come to them. Signals come to those sensors and those get converted in the brain into perceived phenomena. And this process, the science of understanding this, is called perceptual psychology.

It is always advisable to have a little bit of knowledge on perceptual psychology while we are trying to deal with interactive systems in general and extended reality systems in particular. In fact, it is even more important in the context of extended reality systems because here our goal is to explicitly extend our idea of reality which is deeply embedded in our psychology. Now, there are several reasons why we should know perceptual psychology. In fact, these reasons can be understood in the form of some questions. whose answers can come about based on the knowledge of perceptual psychology.

For example, questions like how far away does the object appear to be in a given virtual world? How much video resolution is needed to avoid seeing pixels when we are trying to visualize a virtual world or a virtual object? How many frames per second are enough to perceive motion as continuous? Is the user's head appearing at the proper height in the virtual world so that the point of view is ideally what should be the point of view with respect to the user's height? There can be some more such questions.

Answers to which can be found in perceptual psychology. For example, where is the virtual sound coming from? Why am I feeling nauseated? Why is one experience, why does one experience more tiring effect than another?

Or in other words, why is one experience with a system more tiring as compared to another experience? What is presence? In fact, perceptual psychology can give us a clue to the overall idea of presence. As you can see in the previous set of questions, What we have learned is that perceptual psychology, in fact, can give us some sort of guidelines for building better hardware,

as well as a better understanding of the effects of the system on the user. To answer these questions, what do we need to know under perceptual psychology? Basic physiology of human body including sense organs and neural pathways, how they are connected to the brain. What are the key theories and insights of experimental perceptual psychology? Interference of engineered XR system with our common perceptual processes and resulting implications or side effects.

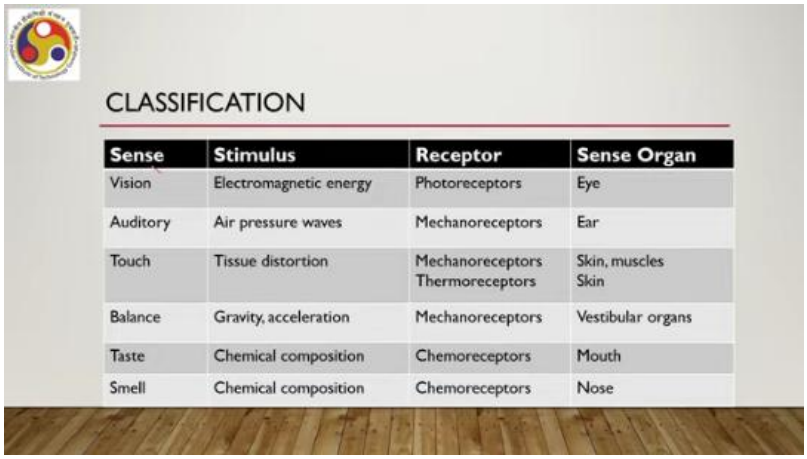
So, these are the things that ideally we should know when we are dealing with interactive systems. Of course, in this lecture, we are not going to get into so much detail; those details can be a course in itself, as you all may know. So, we are going to touch upon a few basic things in perceptual psychology, particularly basic physiology and the insights of how the process of perception works. And also, in subsequent lectures, we will touch upon the third

topic, that is, the interference of engineered XR systems with our common perceptual process.

So, let us first start with our understanding of perceptual physiology. So, we will start by starting with a classification of human senses. What are the senses or sensory organs that we have, and what kind of things are there that allow us to sense? Now, when we are trying to understand sense or sensory organs, sensing in a sense, we should keep in mind that the term sensor

can be used in different contexts. When we are talking about a device or a system, then the term sensor means a device to convert an energy source into electrical signals in a circuit. In the context of the human body, human bodily sensors, on the other hand, refer to an organ, a body organ, to convert a stimulus into neural impulses. Idea-wise similar, structurally different. Now, let us have a quick look at the different sense organs that we have.

So, we have the vision sense; we all know. Now in this case the stimulus is electromagnetic energy that falls on our eye. The receptor that is part of this sense are photoreceptors and the sense organ we all know is the eye. So, when we talk of auditory sense, we have air pressure waves as the stimulus, mechanoreceptors are there and the sense organ is ear. We have then the touch sense.



Tissue distortion is the stimulus in this case. And we have two types of receptors here, mechanoreceptors and thermoreceptors. And the sense organs are skin, muscles. Then we have balance, sense of balance. In this case, the stimuli are gravity or acceleration.

We have mechanoreceptors and we have vestibular organs to act as the sense organs. Taste is there, chemical composition can be the stimulus, chemoreceptors are there and the sense

organ is mouth. Similarly, smell, sense is there, again chemical composition can be the input, the stimulus, chemoreceptors are there and nose is the sense organ. So, we have these six senses out of which Five senses are well known and balance is a sense for which we do not have any specific organ unlike the other five.

So, how sensing works? Now, each receptor is a sensor targeting particular stimuli. So, in the sense organ, we have receptors. Each receptor is a sensor that targets particular stimuli, known as sensory system selectivity.

So, receptors are bodily organs which are designed in a way such that they can get activated only when a specific type of stimulus arrives, and this phenomenon is known as sensory system selectivity. So, as we have seen in the table that we just saw, Our eyes contain photoreceptors. Now, these photoreceptors get activated when electromagnetic energy in the form of visible light falls on our eyes.

So, the photoreceptors are targeting electromagnetic energy in the frequency range of visible light. So, they get activated only in the range of visible light. Auditory, touch, and balance senses involve motion, vibration, or gravitational force sensed by mechanoreceptors. Sense of touch additionally involves thermoreceptors to detect changes in temperature. Our balance sense helps us to know which way our head is oriented or in other words, including sensing the direction of up, the upward direction. Taste and smell, these two senses rely on chemoreceptors, which provide signals based on chemical composition of matter on the tongue or nasal passage, depending on the sense. So, in summary, we can say that we need to have some idea of perceptual psychology in order to understand the role of human factors in extended reality systems. And we also covered a little bit of a part of perceptual psychology that is the physiology of senses.

What are the senses that we deploy to perceive the world around us? We talked about six senses: vision, auditory, balance, taste, smell, and touch. And for each, there are receptors which are part of the sensory organs. The sensory organs are there, and in the organs, there are receptors.

These receptors receive stimuli depending on the organ. So, there are different types of receptors. We have photoreceptors, chemoreceptors, mechanoreceptors, and so on. In the next part of this lecture, we are going to talk about how the sensing process works. So, we have now talked about the hardware of sensing in terms of the human body, that is, the sensory organs.

Next, we will focus on how the hardware helps in perceiving the world. So, see you in the next part of the lecture. I hope you have understood the concepts. Looking forward to meeting you again. Thank you and happy learning.